4. Speech Dataset

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Speech Dataset

1. Introduction
2. Dataset Building
3. Dataset Annotation
4. Data Processing
5. Benchmarking & Evaluation

1. Introduction

- Why we need Speech Datasets?
  - Foundation for training and evaluating speech technology models
  - Main tasks: ASR, TTS, Speaker Recognition, Speaker Diarization, Speech Emotion Recognition, Multimodal Generative AI, Mispronunciation Detection and Diagnosis...

Type of Speech Datasets

- Controlled vs. Spontaneous speech
  - Controlled: in a reading style, clear pronunciation (audio books, reading datasets, etc.)
- Monologue vs. Dialogue vs. Conversational
  - Single, two-, multi-speaker

Type of Speech Datasets

- Scripted vs. improvised
  - For emotion detections
- Clean vs. noisy
  - Some fields cannot be recorded in noise-free environment (air traffic control)

Example: VOiCES Dataset

Popular Speech Datasets

1. ASR

- LibriSpeech, CommonVoice, TEDLIUM, VoxPuli

1. Speaker Recognition & Diarization

- VoxCeleb, DIHARD, AMI Meeting, CNCeleb

1. TTS & Speech Synthesis

- VCTK, LibriTTS

1. Speech Emotion Recognition

- IEMOCAP, MSP-Podcast, MELD

2. Speech Dataset Building

Collection methodologies:

- Recording (expensive, not diverse)
- Crawling (unlabeled/weakly labeled)

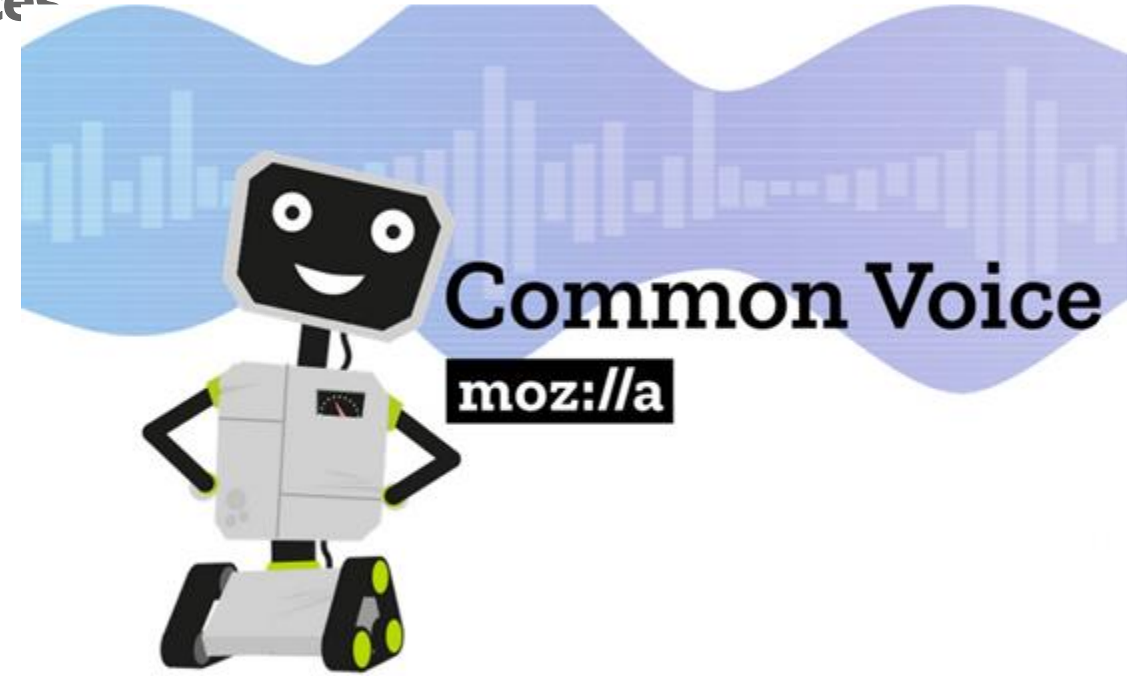
Speech Data Collection

Method	Pros	Cons
Studio recording	- High-quality, clean speech - Controlled conditions (consistent microphone & environment) - Easier transcription & labeling	- Expensive & time-consuming - May not represent real-world conditions - Limited speaker diversity
Crowdsourced recording	- Large-scale collection possible - Speaker diversity - Cheaper than studio recording	- Variable recording quality - Risk of noisy or low-quality samples - Requires extensive quality control

Noise-free Recording in Studio

CommonVoice: Crowd-sourced Recording

Visit https://commonvoice.mozilla.org/ and **donate** some **utterances**



CommonVoice: Crowd-sourced Recording

CommonVoice - more detail

Speech Data Collection

Method	Pros	Cons
Crawling	- Huge amount of data available - Natural, real-world speech - Useful for domain-specific datasets	- Licensing & legal issues - Poor transcription quality (requires re-annotation) - Inconsistent audio quality
Synthesis	- Unlimited data generation - Can simulate rare accents or low-resource languages	- Lack of natural prosody - Can introduce biases - May not generalize well to real-world scenarios

Synthetic Speech Data Generation

- **Step 1:** Define Data Requirements
- **Step 2:** Select a TTS Model for Speech Synthesis
- **Step 3:** Prepare Text Data for Speech Synthesis
- **Step 4:** Generate Synthetic Speech Data
- **Step 5:** Post-Processing & Data Augmentation

Step 1: Define Data Requirements

- Which task?
- Diverse in terms of speaker, speed, and intonation?
- How many hours of data are required?

Task	Minimum Hours	Optimal Hours	Diversity Requirements
ASR (Speech-to-Text)	100-500h	5,000-10,000h	Multiple speakers, accents, background noise, different speaking speeds
Voice Cloning (1-shot/few-shot)	1-5h per speaker	10-20h per speaker	High-quality recordings, same microphone conditions, same speaking style
Text-to-Speech (TTS) Training	10-50h	500-1000h	Multiple voices, different emotions, balanced phoneme distribution
Speaker Recognition	1-5h per speaker	500-1000h	Various microphones, environments, speaking styles
Emotion Recognition	50-100h	500-1000h	Different emotional tones, gender balance, age diversity

Step 2: TTS Models for Speech Synthesis

- **FastSpeech 2, Tacotron 2 + WaveGlow/WaveRNN:** Natural-sounding speech but requires fine-tuning.
- **VITS (End-to-End Vocoder + TTS):** High-quality speech with fast synthesis speed.
- **F5-TTS:** multi-speaker & multilingual, high-quality, expressive , but requires fine-tuning for best quality.
- **Neural TTS API (Microsoft Azure, Google Cloud, Amazon Polly):** Fast synthesis, high-quality speech but limited customization.

TTS Models for Speech Synthesis

Model	Key Features	Advantages	Disadvantages
FastSpeech 2, Tacotron 2 + WaveGlow/WaveRNN	Natural-sounding speech, prosody modeling	**High-quality, expressive speech**	Requires **fine-tuning** for optimal performance
VITS (End-to-End Vocoder + TTS)	GAN-based architecture, end-to-end synthesis	**High-quality speech with fast synthesis speed**	May require **high computational power**
F5-TTS	Multi-speaker & multilingual, expressive speech	**High-quality, expressive speech, adaptable for multiple languages**	Requires **fine-tuning** for best results
Neural TTS API (Microsoft Azure, Google Cloud, Amazon Polly)	Pre-trained commercial API, optimized synthesis	**Fast synthesis, high-quality speech, low latency**	**Limited customization,** predefined voices

Step 3: Prepare Text Data for Speech Synthesis

- **Ensure a diverse text corpus** (covering phonemes, vocabulary, and sentence structures)
- **Use proper punctuation and spacing** to improve synthesis quality
- **Balance content** to avoid dataset bias
- **Apply text augmentation** techniques to increase diversity

Best practices for text corpus selection:

- Sentences covering different tonal variations (for tonal languages like Vietnamese)
- A mix of short, long, declarative, and interrogative sentences
- Dialogues, news texts, audiobook transcripts, or movie scripts

Step 4: Generate Synthetic Speech Data

- Run **TTS synthesis** to generate **WAV (or FLAC) files**.
- **Combine multiple voices** to create a more balanced dataset.

Example for SER task:

- Use an expressive TTS model (e.g., F5-TTS, VITS, or StyleTTS2) that supports emotion synthesis.
- Balance the dataset so that each emotion is equally represented.

Step 5: Post-Processing & Data Augmentation

- **Add background noise**: Helps ASR models handle real-world conditions
- **Modify speech rate, pitch, and emotion:** Improves robustness to different speaking styles
- **Simulate different microphone effects:** Improves model generalization

Useful tools for augmentation:

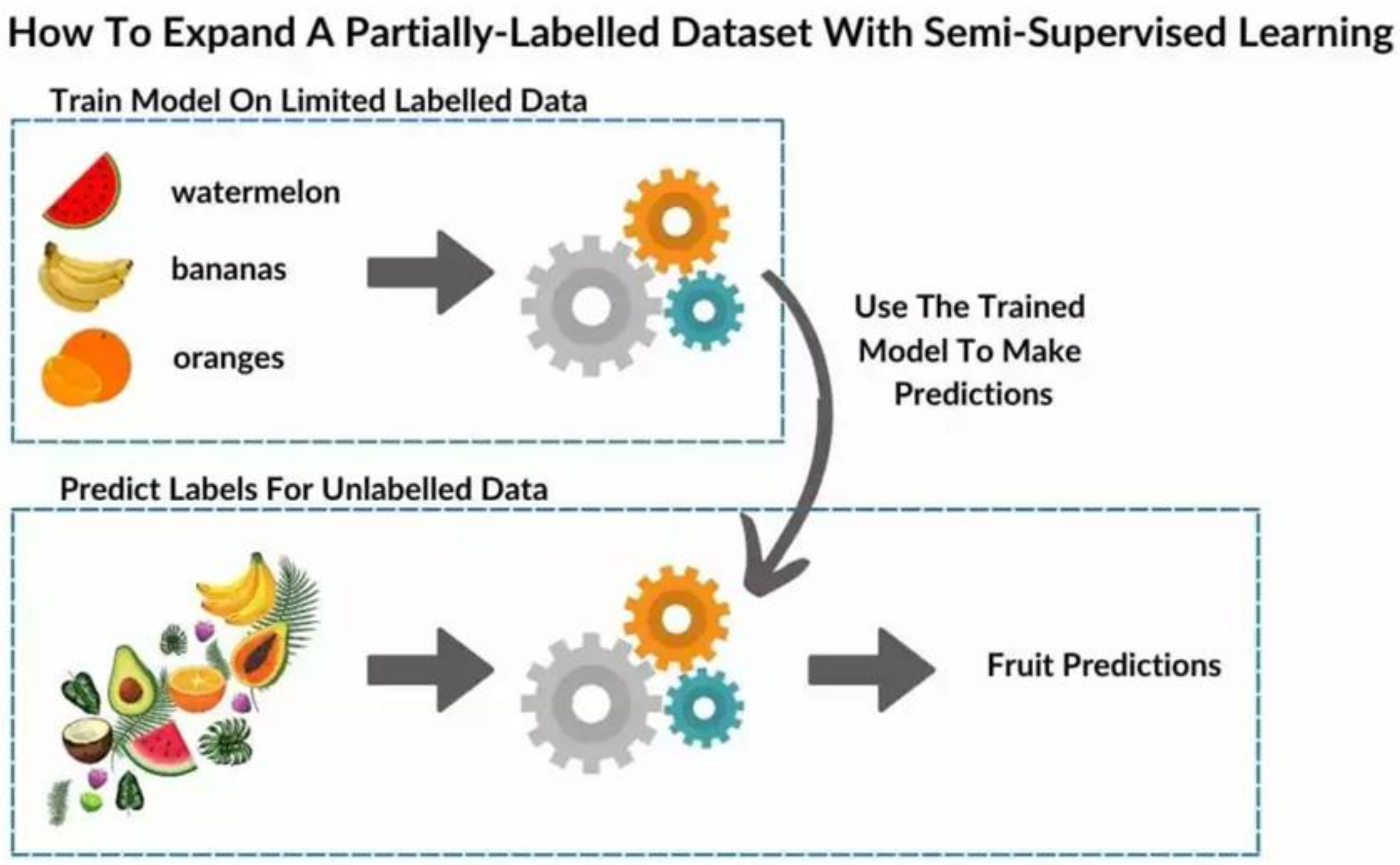
- **SoX, ffmpeg** for modifying speech characteristics
- **Room Impulse Response (RIR)** for simulating recording environments
- **WaveAugment, SpecAugment** for perturbing audio signals.

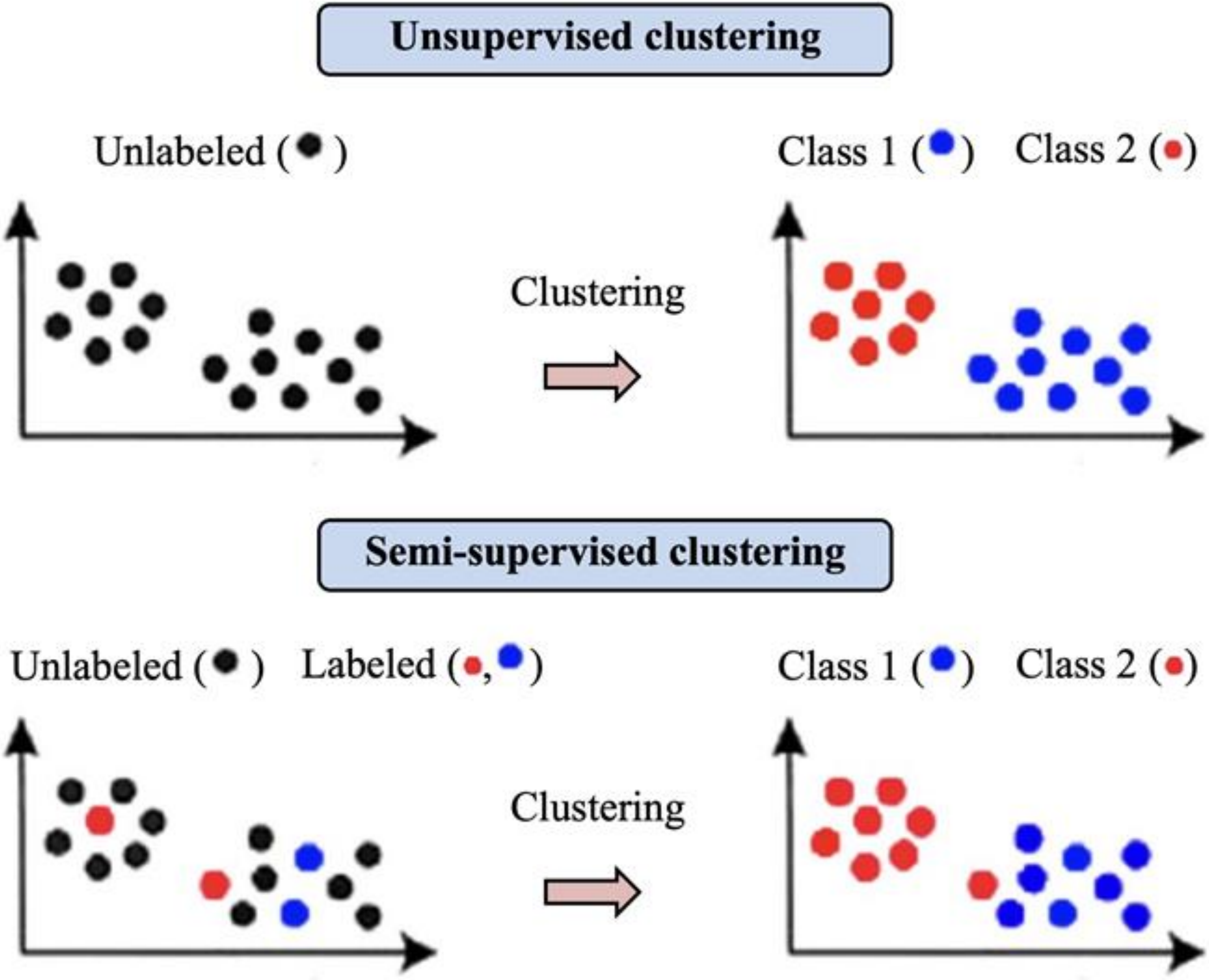
Step 5: Example

- **ASR:**
  - Adding **background noise** (e.g., cafe noise, street noise, reverberation).
  - Applying **speed perturbation** (slightly speeding up or slowing down speech).
  - Mixing **low-quality and high-quality samples** (e.g., phone call audio vs. studio recordings).
- **Voice Cloning**
  - **Speaking speed** (to simulate different conversational styles).
  - **Pitch & intensity** (to add natural variation).
- **SER:** Generate speech samples with different emotions, ensuring:
  - Variations in pitch, intensity, and speaking rate.
  - Different speakers expressing the same emotion (to prevent speaker bias).

3. Speech Data Annotation

- Human annotation: more correct, but requires large-scale quality control, and expensive
- Weak labels: mostly generated by larger models of the same task
  - Auto-generated transcriptions as labels for ASR models (Whisper)
  - Clustering for speaker labels
- Semi-supervised annotations

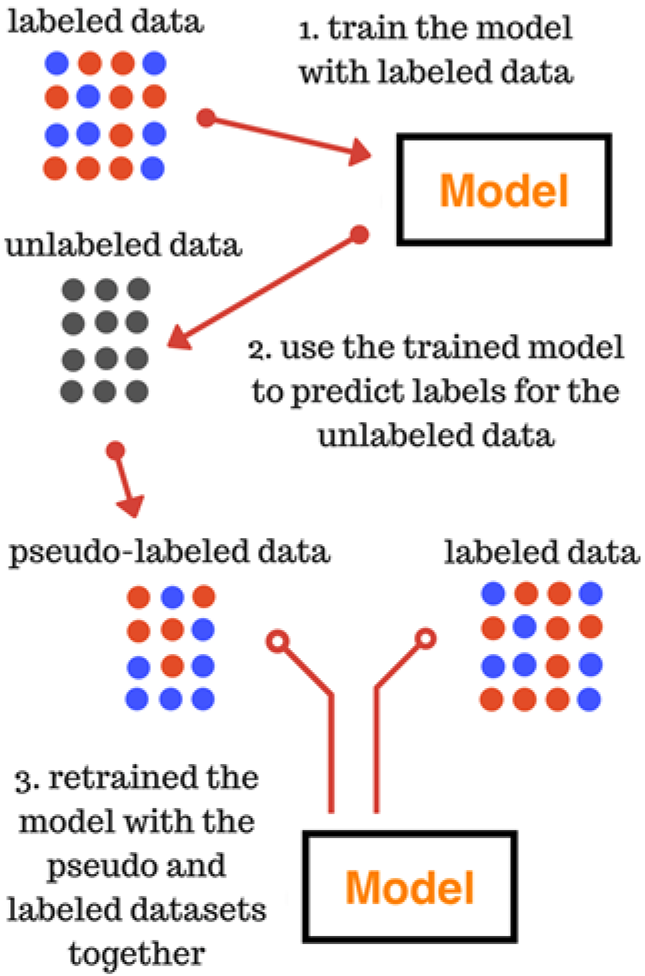

How To Expand A Partially-Labelled Dataset With Semi-Supervised Learning
Train Model On Limited Labelled Data
watermelon
bananas
oranges
Use The Trained Model To Make Predictions
Predict Labels For Unlabelled Data
Fruit Predictions



Shi, X., Yue, C., Quan, M. *et al.* A semi-supervised ensemble clustering algorithm for discovering relationships between different diseases by extracting cell-to-cell biological communications. *J Cancer Res Clin Oncol* 150, 3 (2024)

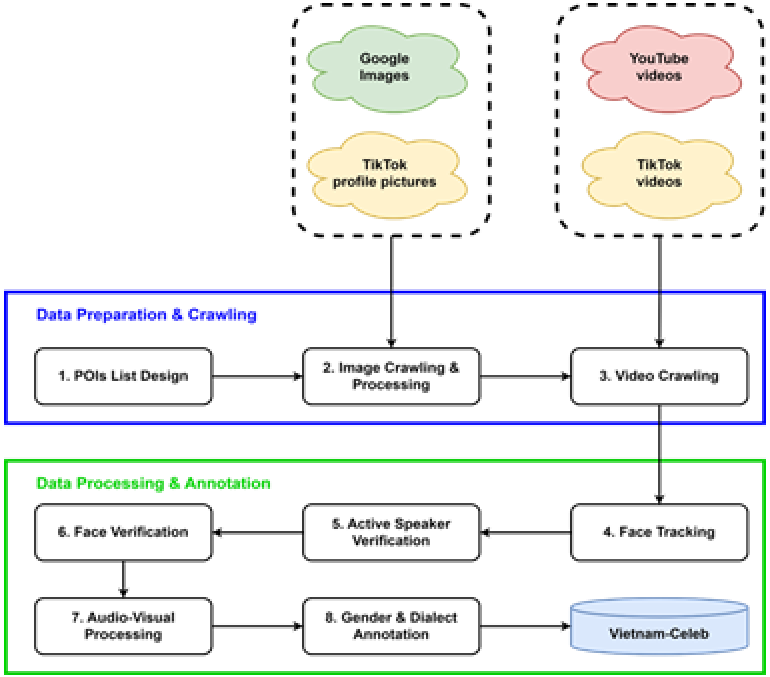
Pseudo Labeling

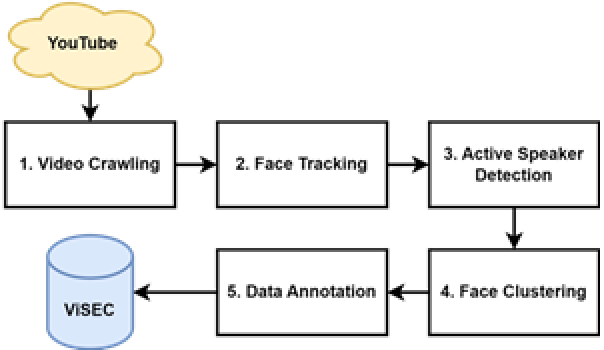
1. Train a model with labelled data
2. Use the trained model to predict labels for the unlabeled data, which creates pseudo-labeled data.
3. Retrain the model with the pseudo-labeled and labeled data together.



Example: Vietnam-Celeb PIPELINE

- Crawling from public media
  - Visual-audio construction pipeline
    - 100% automatic or semi-auto labeling
  - Tasks: Speaker Recognition, SER
- Recording in spontaneous environments
  - Crowded sourcing with validation process or expert annotation
  - Tasks: MD&D, SLU





(Pham et al.) Pham, V.T., Nguyen, X.T.H., Hoang, V., Nguyen, T.T.T. (2023) Vietnam-Celeb: a large-scale dataset for Vietnamese speaker recognition. Proc. INTERSPEECH 2023, 1918-1922, doi: 10.21437/Interspeech.2023-1989

Visual-aid for Speaker Identification

Face Tracking

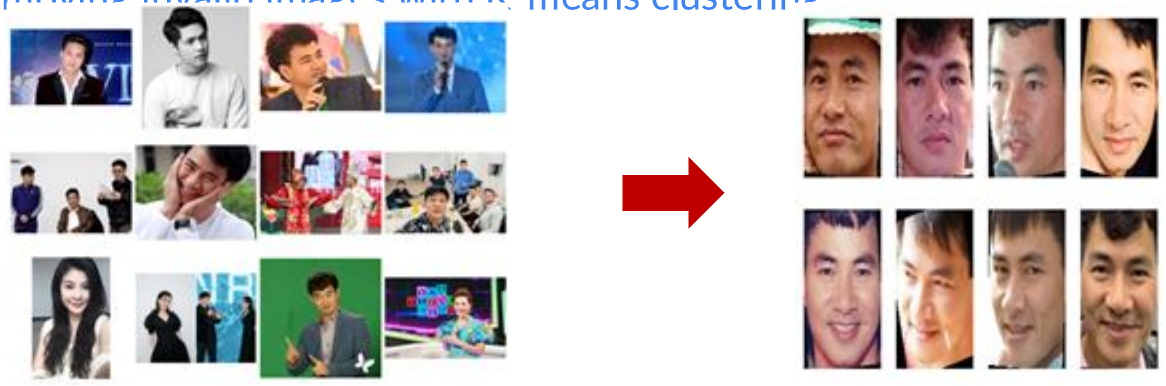
- With the crawled videos, face tracking is performed to extract videos of only facial areas.
- Use the S3FD (Zhang et al. 2017) model for detecting faces in a video. This is a model that can run in real-time and can detect faces of various sizes and scales.
- **Face stream videos are produced by aggregating consecutive facial bounding boxes where the Intersection over Union (IOU) score is higher than 0.5.**

(Zhang et al. 20217) Zhang, S., Zhu, X., Lei, Z., Shi, H., Wang, X., & Li, S. Z. (2017). S3fd: Single shot scale-invariant face detector. In Proceedings of the IEEE international conference on computer vision (pp. 192-201).

Visual-aid for Speaker Identification

Face Detection

- Performing face detection using RetinaFace (Deng et al. 2020)
- Extracting face embeddings / representations using ArcFace (Deng et al. 2019)
- Removing invalid images with K-means clustering



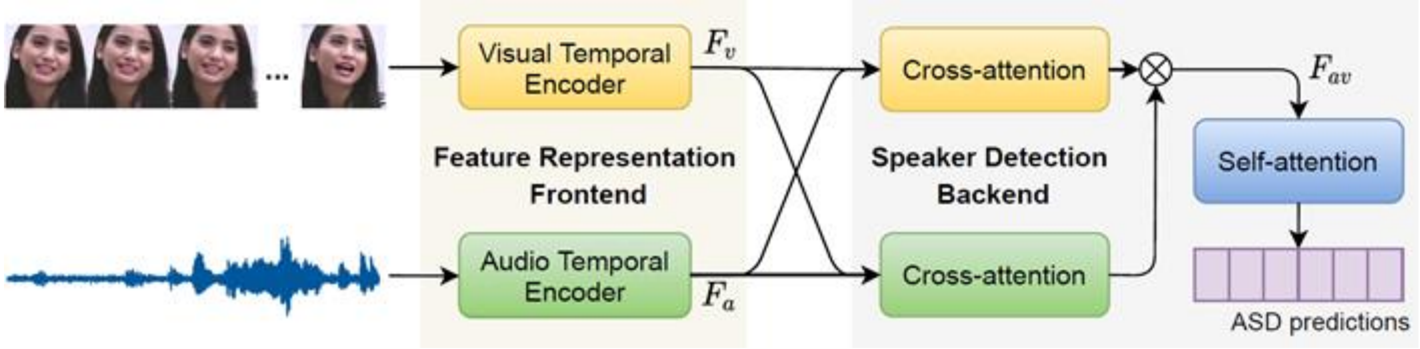
Xuân Bắc
(actor/comedian)

(Deng et al. 2020) Deng, J., Guo, J., Ververas, E., Kotsia, I., & Zafeiriou, S. (2020). Retinaface: Single-shot multi-level face localisation in the wild. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 5203-5212).
(Deng et al. 2019) Deng, J., Guo, J., Xue, N., & Zafeiriou, S. (2019). Arcface: Additive angular margin loss for deep face recognition. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 4690-4699).

Visual-aid for Speaker Identification

Active Speaker Verification

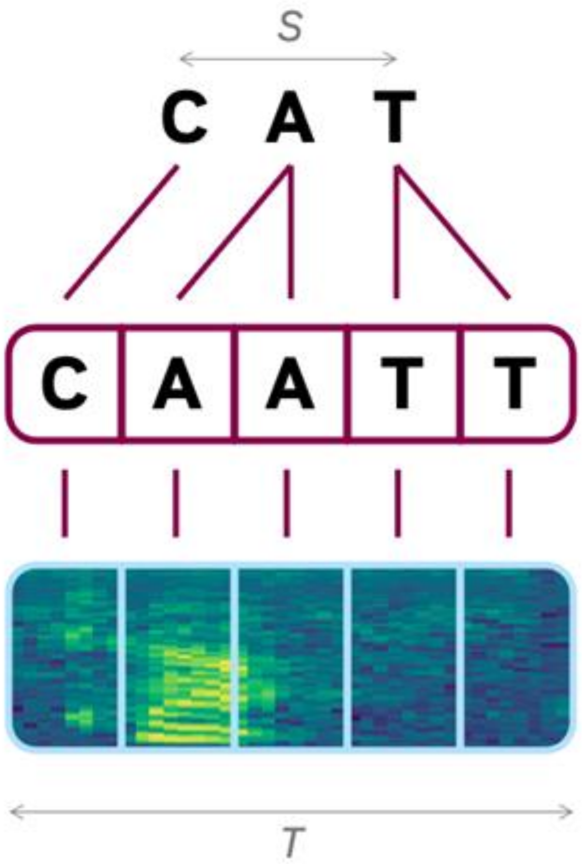
- Use the TalkNet (Tao et al. 2021) model to verify if the person in a video is really speaking
- For each video input, TalkNet outputs a **sequence of confidence scores** corresponding to the frames of the video.
- This sequence of scores is then used to mark out the videos where the person is actually speaking.



(Tao et al. 2021) Tao, R., Pan, Z., Das, R. K., Qian, X., Shou, M. Z., & Li, H. (2021, October). Is someone speaking? exploring long-term temporal features for audio-visual active speaker detection. In Proceedings of the 29th ACM International Conference on Multimedia (pp. 3927-3935).

Introduction

- Process of **automatically aligning speech audio with its corresponding phoneme, word, or sentence-level transcription**
  - Determines **the precise timestamps** (start and end times) of words or phonemes in a given speech recording.



Why we need forced alignment?

- **Speech Dataset Annotation**: Helps align transcriptions with speech data, enabling better supervised learning for ASR, TTS… models
- **Phonetic and Prosody Analysis** – Used in linguistic research to analyze pronunciation, stress, and intonation.

Common Alignment methods

Method	Example Tools	Strengths	Weaknesses
(A) HMM-GMM Based Forced Alignment	HTK, Kaldi, MFA (GMM mode)	**Precise phoneme alignment**, time-proven	Requires lexicons, struggles with noisy speech
(B) Deep Learning-Based Forced Alignment	MFA (DNN mode), Gentle, Deep Phonetics Aligners	**More robust to speaker variations and accents**	Requires large datasets for training
(C) CTC-Based Forced Alignment (Hybrid)	Kaldi (Hybrid HMM-DNN), DeepSpeech, Wav2Vec 2.0	**Faster than HMM, does not require frame alignment during training**	Less precise than HMM for phoneme-level segmentation
(D) End-to-End ASR-Based Forced Alignment	Whisper, Aeneas, DeepSpeech	**Handles real-world speech well**, good for word alignment	Lacks precise phoneme-level segmentation

HMM-GMM-based Forced Alignment

- Traditional method based on **acoustic models trained with HMM-GMM (Gaussian Mixture Models)**.
- Works by **aligning phonemes to the most probable time frames** based on the model's acoustic scores.
- **Used in:** CMU Sphinx, Kaldi, HTK.
- **Limitation:** Less accurate for spontaneous speech due to **rigid phoneme modeling**.

Deep Learning-Based Forced Alignment

- Uses **Deep Neural Networks (DNNs), Convolutional Neural Networks (CNNs), or Transformers** for better acoustic modeling.
- More robust to **speaker variations, accents, and background noise**.
- **Used in:** MFA (Montreal Forced Aligner), Gentle (Kaldi-based).
- **Limitation:** Requires **more computational resources** compared to HMM-based methods.

CTC-Based Forced Alignment (Hybrid)

- Uses **Connectionist Temporal Classification (CTC)** to align speech with text **by optimizing over possible paths, allowing for flexible time alignment at the word or phoneme level rather than requiring strict frame-wise labeling.**
- Can handle **variable-length speech**, making it **faster than HMM-based alignment** while maintaining reasonable accuracy.
- Works well for **phoneme, word, or character-level alignment** depending on the task.
- Used in: **Kaldi (Hybrid HMM-DNN mode), DeepSpeech, Wav2Vec 2.0**.
- **Limitation:** Less precise than **HMM-based forced alignment**, especially for fine-grained phoneme boundaries.

End-to-End ASR-based Forced Alignment

- Uses **state-of-the-art ASR models (e.g., Whisper)** for alignment.
- Can handle **spontaneous speech, conversational speech, and noisy environments**.
- **Used in:** Aeneas, Whisper-based Forced Alignment.
- **Limitation:** Can be **less precise at phoneme-level alignment** compared to dedicated phonetic models.

Common Tools for Forced Alignment

Tool	Methodology	Features	Use Cases
Montreal Forced Aligner (MFA)	HMM + Deep Learning	Supports multiple languages, high accuracy, word/phoneme-level alignment	ASR dataset preparation, phonetic analysis
Gentle Forced Aligner	Kaldi-based	Robust for English, easy to use, handles noisy data	ASR pre-processing, word alignment
Aeneas	ASR-based (DeepSpeech, Kaldi)	Aligns text with speech at word/sentence level	Audiobooks, TTS alignment
Prosodylab-Aligner	HMM-GMM	Lightweight, works well with clean speech	Linguistic research, phoneme alignment
HTK (Hidden Markov Toolkit)	HMM-based	Older but widely used in speech research	ASR training, phonetic segmentation
Whisper Forced Aligner	Transformer-based ASR	High accuracy, zero-shot alignment	Automatic transcription, ASR tuning

4. Speech Data Processing

- Clean and well-segmented datasets improve the accuracy of speech recognition, speaker identification, and other AI-driven speech applications.
- Speech augmentation for more variances

Speech Processing

- **Speaker Diarization:** Identify and label different speakers in conversations.
- **Speech Segmentation:** Split long recordings into manageable utterances (using VAD)
- **Noise Reduction:** Remove background noise and reverberation if needed.
- **Volume Normalization:** Standardize audio levels for consistency.
- **Resampling**: Convert all recordings to a common sample rate (e.g., 16kHz for ASR).
- **Silence Trimming:** Remove excessive silence at the beginning and end.

Speech Data Augmentation

- **Noise augmentation, room simulation**
  - Background noise addition
  - Simulating reverberation effects with room impulse responses
- **Speed perturbation:** changing playback speed while preserving pitch
- **SpecAugment**: masking parts of the spectrograms

Noise Reduction in Datasets Preprocessing

- Unwanted sounds that interfere with the clarity and intelligibility of speech
- Noise reduces the accuracy of speech recognition and speaker verification models.
- Helps in creating a standardized, high-quality dataset for training AI models.

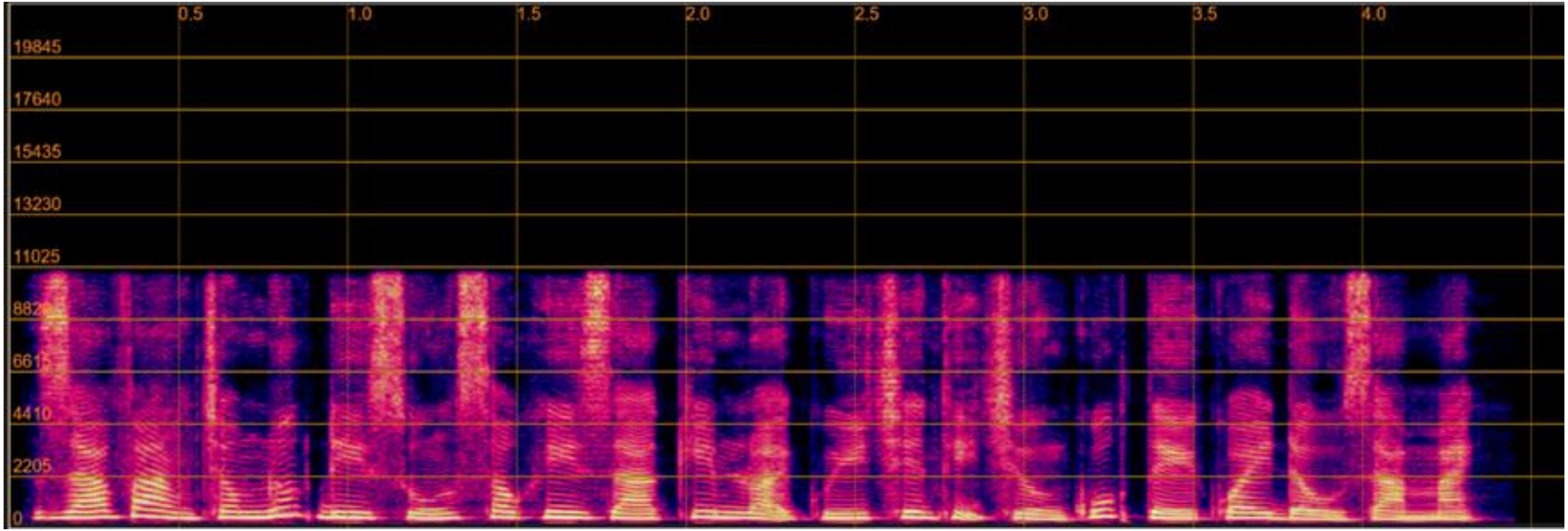
Types of Noises in Speech Datasets

- **Background disturbances in recordings**:
    - Environmental sounds (traffic, fan noise, crowd chatter).
    - Microphone artifacts (hissing, electrical hum).
- **Transient Noise:** Short-duration sounds, including clicks, pops, or microphone handling noises.
- **Reverberation:** Reflections of the speech signal off surfaces, causing a prolonged and muffled sound.

Types of Noises in Speech Datasets

Clean audio examples:

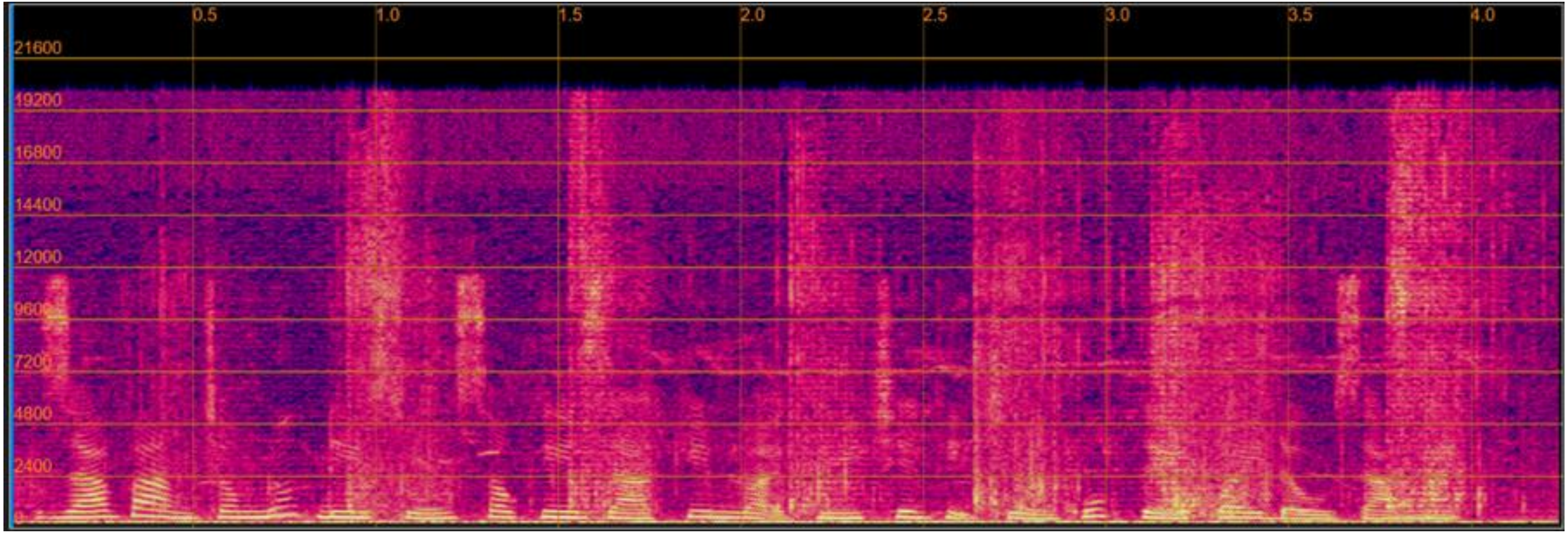
Melspectrogram:



Types of Noises in Speech Datasets

Audio with background noise examples:

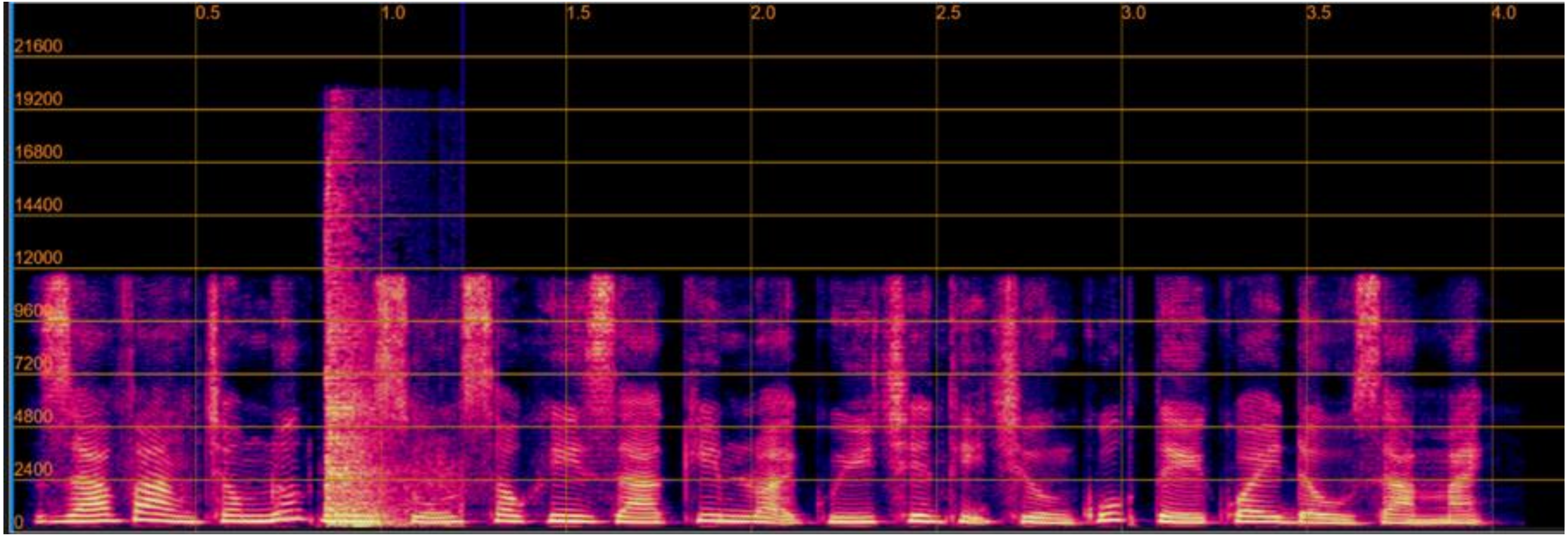
Melspectrogram:



Types of Noises in Speech Datasets

Audio with transient noise examples:

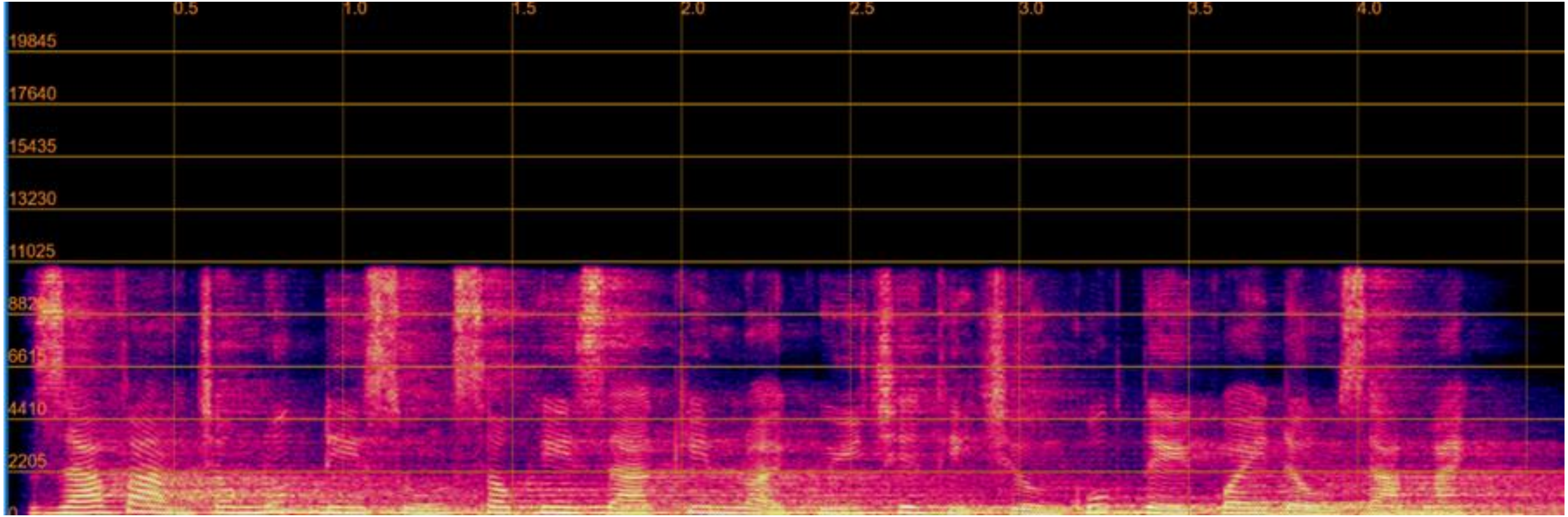
Melspectrogram:



Types of Noises in Speech Datasets

Reverberation audio examples:

Melspectrogram:



Noise Removal Techniques

- **Spectral Subtraction** – Identifies and subtracts background noise from speech signals.
- **Adaptive Filtering** – Uses algorithms like Wiener filters to suppress unwanted noise.
- **Deep Learning-based Denoising** – Neural networks (e.g., U-Net, Denoising Autoencoders) trained to filter out noise.

Silence Trimming in Datasets Preprocessing

What is Silence in Speech Data?

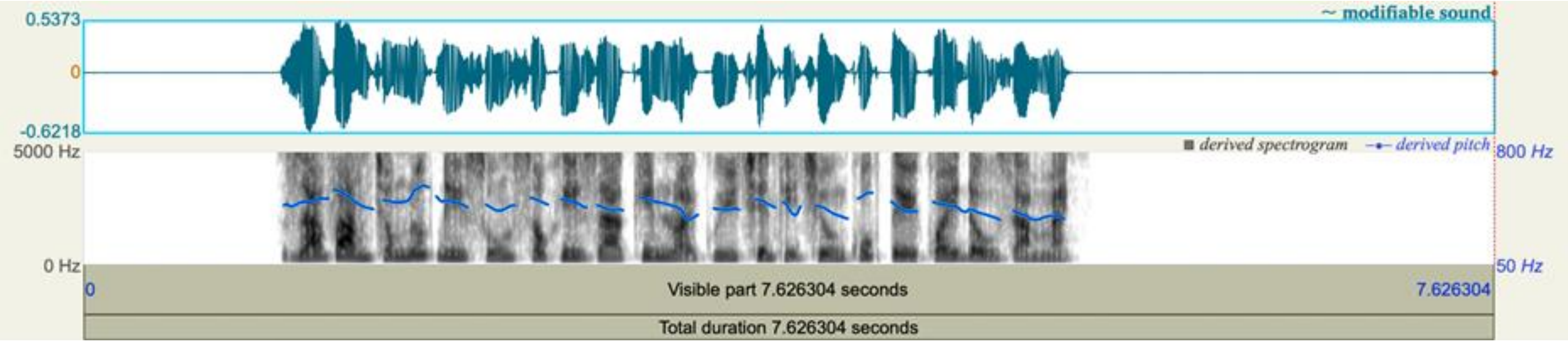
- **Intentional Silence**: Pause between words/sentences
- **Unintentional Silence**: Long gaps due to recording errors, microphone delay, or background interference.

Why Trim Silence?

- Reduces dataset size and training time.
- Prevents silent frames from biasing the model.
- Makes speech recognition models more responsive.

Silence Trimming Examples

	Before trimming	After trimming
Duration	7s	4s
Size	660Kb	400Kb



Silence Trimming Techniques

- **Energy-Based Detection**: Removes frames with very low amplitude.
- **Thresholding Methods**: Sets a dB threshold to define silence.
- **Voice Activity Detection (VAD)**: Machine learning-based detection of speech vs. silence.

Practical Tools for Silence Trimming

Pydub

librosa

FFmpeg

Speech Segmentation in Datasets Preprocessing

- Speech segmentation is the process of dividing continuous speech into distinct, meaningful units such as sentences, words, or phonemes.
- This process is essential in various speech-related applications, including automatic speech recognition (ASR), speaker diarization, and language modeling.
- Effective segmentation enhances the performance of these applications by providing structured and analyzable speech data.

Applications of Speech Segmentation

- **Automatic Speech Recognition (ASR):** Accurate segmentation aids in identifying word boundaries, improving transcription quality.
- **Speaker Diarization:** Determines "who spoke when," facilitating the analysis of multi-speaker recordings.
- **Language Modeling:** Segmentation into smaller units allows for the study of linguistic patterns and the development of more robust language models.

Speech Segmentation Techniques

- **Voice Activity Detection (VAD)**: Distinguishes between speech and non-speech segments, focusing processing on relevant data.
- **Speaker Diarization**: Partitions audio based on speaker identity, essential for multi-speaker environments.
- **Phoneme Segmentation**: Breaks down speech into individual phonetic units, crucial for phonetic analysis and speech synthesis.

Voice Activity Detection (VAD)

Purpose: Identifies segments containing speech, filtering out silence and background noise.

Methods:

- **Energy-Based Approaches**: Analyze signal energy to detect speech presence.
- **Machine Learning Approaches**: Utilize trained models to differentiate speech from non-speech segments.

Speaker Diarization

Definition: The process of partitioning an audio stream into homogeneous segments according to the identity of each speaker.

Techniques:

- **Clustering Algorithms:** Group similar voice characteristics to identify distinct speakers.
- **Neural Network-Based Methods:** Employ deep learning models for more accurate speaker identification.

Applications: Useful in transcribing meetings, interviews, and analyzing conversational dynamics.

5. Benchmarking & Evaluation

- Large-scale speech datasets are crucial for ASR, TTS, Speaker Profiling, and Voice Cloning.
- Poor data quality can lead to biased or inaccurate models.
- Key evaluation criteria: Diversity, Quality, Balance, Labeling Accuracy

Data Diversity & Representativeness

- Ensuring that the dataset covers a **wide range of speakers**, **accents**, and **speaking conditions** is crucial.
- **Metrics**:
  - **Speaker Count & Distribution:** Ensure a balanced dataset across different genders, age groups, and accents.
  - **Phonetic Coverage**: Measure how well phonemes in the dataset match real-world speech patterns.
  - **Linguistic Coverage**: Compute word frequency distributions to detect bias toward certain vocabulary or sentence structures.
  - **Environment Variety**: Analyze different acoustic conditions (e.g., background noise levels, room reverberation).

Data Quality & Cleanliness

- Low-quality or noisy speech data can negatively impact model performance, making quality assessment crucial.
- **Metrics**:
  - **Signal-to-Noise Ratio (SNR)**: Measures how much speech is present relative to background noise. (Higher SNR = better quality).
  - **Perceptual Evaluation of Speech Quality (PESQ)**: Quantifies speech quality as perceived by humans.
  - **Short-Time Objective Intelligibility (STOI)**: Measures how understandable speech is, especially in noisy environments.
  - **Word-Level and Phoneme-Level Annotation Accuracy**: Checks alignment errors in labeled datasets.

Dataset Balance & Bias Analysis

- An imbalanced dataset can cause bias in models, leading to poor generalization.
- Metrics:
  - **Gender & Age Distribution**: Ensure dataset has fair representation across different demographics.
  - **Accent & Language Distribution**: Check if the dataset is biased toward a specific accent or dialect.
  - **Outlier Analysis**: Identify if certain speaker groups are underrepresented.

Labeling Accuracy & Annotation Quality

- For supervised learning, high-quality labels are essential. Misaligned or noisy labels can degrade performance.
- **Metrics:**
  - **Labeling Consistency Score**: Measures variation in transcription across annotators.
  - **Alignment Accuracy (ASR datasets):** Checks if word/phoneme timestamps match actual speech.
  - **Inter-Annotator Agreement (IAA):** (Cohen's Kappa Score, Fleiss' Kappa) – Measures how much agreement exists between human annotators.
- A recent trend: LLM as a judge

Best practices for Dataset Evaluation

- **Use Automated Tools for Quality Checks**
  - Noisiness & Speech Quality → SoX, librosa, PESQ, SNR estimators
  - Diversity Analysis → t-SNE visualization of speaker embeddings
  - Bias Detection → Gender, Age, Accent Distribution Analysis
- **Run Benchmark Models on Subsets**
  - Evaluate Word Error Rate (WER), Character Error Rate (CER), and Speaker Verification EER before using the full dataset.

Best practices for Dataset Evaluation

- **Human-in-the-Loop Validation**
  - Use a small subset for manual annotation verification.
- **Monitor Dataset Drift**
  - If adding new data, check for shifts in distribution (e.g., changes in speaker demographics, recording quality).